

ATVB IN FOCUS:
Preeclampsia

Series Editor: Jessica Faulkner

Renin-Angiotensin-Aldosterone System in Preeclampsia: Pathophysiological Insights and Links to Vascular Dementia

Jenny Lutshumba¹, Elizabeth Mirsky², Kelsey A. Bullens³, Emma G. Smith, Esther L. Ndashaala, Robin Finch⁴

ABSTRACT: Preeclampsia, a condition characterized by new-onset hypertension during pregnancy and ≥ 1 indices of organ damage, is a major driver of both short- and long-term maternal health outcomes. Individuals with a history of preeclampsia are at increased risk for cardiovascular, renal, and other chronic diseases. Recent studies have linked preeclampsia history with increased risk of dementia later in life, underscoring the importance of new guidelines emphasizing blood pressure control to reduce the risk of cognitive decline. Unfortunately, neither the mechanisms underlying the causes of preeclampsia nor their links with cognitive function are well understood. Vascular contributions to cognitive impairment and dementia are a type of dementia caused by reduced cerebral blood flow, often resulting from vascular dysfunction associated with small vessel damage or chronic hypertension. The renin-angiotensin-aldosterone system (RAAS) is a hormone system with a major role in regulating blood pressure, fluid homeostasis, and vascular function inside and outside of pregnancy. The RAAS is markedly activated during pregnancy; preeclampsia is associated with disruptions in the normal activity of the RAAS, including excessive Ang (angiotensin) II type 1 receptor signaling, and loss of protective effects of Ang-(1-7) and Ang II type 2 receptors. Preexisting hypertension may impair the normal RAAS response to pregnancy, increasing susceptibility to vascular damage both during and after pregnancy. Results from human and animal studies indicate that dysregulation of the RAAS is a shared pathway underlying vascular dysfunction in both preeclampsia and vascular contributions to cognitive impairment and dementia, suggesting a mechanistic link between these 2 conditions. Persistent endothelial damage, impaired vascular remodeling, and chronic activation of Ang II type 1 and mineralocorticoid receptor signaling may increase long-term risk. This connection underscores the importance of monitoring and managing blood pressure during pregnancy and in women with a history of preeclampsia to mitigate the risk of dementia.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: aging ■ angiotensins ■ cerebral small vessel disease ■ cognitive dysfunction ■ endothelium ■ hypertension ■ pregnancy

Preeclampsia is a serious condition presenting after 20 weeks of gestation characterized by new-onset elevated blood pressure and organ damage, and widespread endothelial dysfunction.¹ Preeclampsia is accepted to be a disease of placental origins progressing in 2 stages: defective or abnormal placentation in early pregnancy, followed by manifestation of the maternal syndrome in the second half of pregnancy² (depicted in Figure 1). Clinical diagnosis includes systolic/diastolic blood pressure $>140/90$ mm Hg (2 measurements over

4 hours or a single measurement of $>160/110$ mm Hg plus 1 index of organ damage (Figure 1).

The pathogenesis of preeclampsia in the first stage includes abnormal remodeling of the uterine spiral arteries, ultimately resulting in poor placental perfusion and intermittent hypoxia/reperfusion, release of circulating antiangiogenic factors, production of inflammatory

Please see www.ahajournals.org/atvb/atvb-focus for all articles published in this series.

Correspondence to: Robin Finch, PhD, Division of Pediatric Research, Department of Pediatrics, University of Kentucky, 138 Leader Ave, Lexington, KY 40508. Email robin.finch@uky.edu

© 2026 American Heart Association, Inc.

Arterioscler Thromb Vasc Biol is available at www.ahajournals.org/journal/atvb

Nonstandard Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
ADAM17	ADAM metallopeptidase domain 17
ADH	antidiuretic hormone
AGT	angiotensinogen
Ang	angiotensin
AT1R	angiotensin II type 1 receptor
AT1R-AA	autoantibodies to the angiotensin II type 1 receptor
AT2R	angiotensin II type 2 receptor
BBB	blood-brain barrier
eNOS	endothelial nitric oxide synthase
MR	mineralocorticoid receptor
PIGF	placental growth factor
RAAS	renin-angiotensin-aldosterone system
sFlt-1	soluble fms-like tyrosine kinase-1
VCID	vascular contributions to cognitive impairment and dementia
VEGF	vascular endothelial growth factor

Highlights

- Preeclampsia is linked with acute and long-lasting vascular dysfunction, which increases long-term risk for hypertension and vascular contributions to cognitive impairment and dementia.
- Dysregulation of the renin-angiotensin-aldosterone system is a shared mechanistic pathway linking preeclampsia to vascular contributions to cognitive impairment and dementia.
- Excess angiotensin II type 1 receptor and mineralocorticoid receptor signaling, with loss of local protective renin-angiotensin-aldosterone system factors, impairs the normotensive adaptation to pregnancy and results in endothelial and organ damage.
- Preexisting or pregnancy-induced hypertension may amplify maladaptive renin-angiotensin-aldosterone system responses, increasing susceptibility to lasting cerebrovascular damage.
- Pregnancy represents a critical window for identifying and modifying vascular risk to improve long-term cardiovascular and brain health in women.

cytokines, and oxidative stress at the placenta.³ These factors contribute to the onset of the maternal syndrome, characterized by increasing systemic vascular resistance and widespread endothelial dysfunction that affects multiple organs.⁴ Both the initial development and the manifestation of the maternal syndrome are influenced by complex factors: immunologic (self-response at the placenta and other immune cell alterations), genetic (polymorphisms in genes regulating renal function and fluid homeostasis),² and acquired (prepregnancy vascular dysfunction or history of acute kidney injury).⁵

Preeclampsia and hypertensive disorders of pregnancy affect an estimated 7.5% and 15.3%, respectively, of women in their reproductive years.⁶ Rates of hypertension in pregnancy are increasing, likely reflecting population-level increases in obesity and cardiometabolic risk factors, as well as increasing maternal age during pregnancy.⁷ Risk factors for preeclampsia include advanced maternal age (>35 years of age), nulliparity (first-time pregnancies have a higher risk), maternal obesity, Black women, multiple gestation, and preexisting medical conditions such as a history of hypertension, diabetes, autoimmune, or renal disease.⁸ Adverse outcomes of preeclampsia include hemolysis, elevated liver enzymes, and low platelets syndrome, acute renal or liver failure, placental abruption, preterm labor/birth (spontaneous or medically indicated), stroke, pulmonary edema, intrauterine growth restriction, postpartum hemorrhage, postpartum preeclampsia, and maternal and fetal death.^{1,5,9} Chronic (prepregnancy) hypertension particularly increases the risk for both preeclampsia and associated adverse outcomes, underscored by a recent population study, where

the risk for preeclampsia was increased 6-fold, and the risk for preterm birth or fetal growth restriction was increased 3-fold.¹⁰ In low- and middle-income countries, preeclampsia is a leading cause of maternal and perinatal mortality, responsible for 14% of global maternal deaths¹¹ and 7.8% of maternal deaths in the United States, where maternal deaths remain higher than those of other high-income countries.¹² Inequities in access to health care exacerbate disparities, with limited prenatal screening and management contributing to poor outcomes.¹³

Preeclampsia also increases the risk for subsequent maternal cardiovascular disease. Numerous epidemiological studies indicate that women with a history of preeclampsia are at increased risk for coronary heart disease and heart failure,¹⁴ stroke,¹⁵ hypertension,¹⁶ and death from cardiovascular disease¹⁷ as early as 1 to 3 years following delivery. Guidelines by the American Heart Association recommend recognizing preeclampsia as a cardiovascular risk factor.^{18,19} Explanations for the association between these conditions include shared risk factors among preeclampsia and cardiovascular diseases, such as obesity and metabolic disorders, or the presence of latent conditions that increase cardiovascular vulnerability, which are revealed during the stress test of pregnancy. Independent of preexisting risk, preeclampsia may have lasting effects on the vasculature that further increase susceptibility over time.²⁰

HEALTHY PREGNANCY AND THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

During pregnancy, the maternal cardiovascular system undergoes significant adaptations to meet the

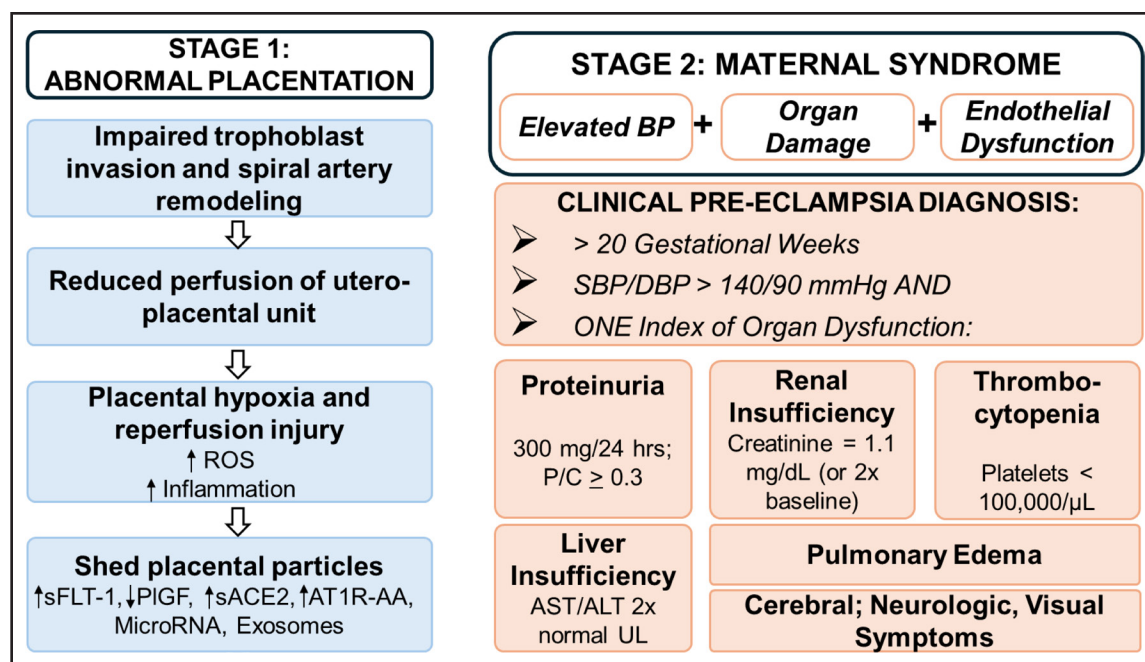


Figure 1. Development of preeclampsia and clinical diagnosis.

Stage 1 (abnormal placentation) and stage 2 (manifestation of maternal syndrome) of preeclampsia. Clinical diagnosis is made in stage 2 based on new-onset elevated blood pressure after 20 weeks of gestational age plus 1 index of organ dysfunction. ALT indicates alanine aminotransferase; AST, aspartate aminotransferase; AT1R-AA, angiotensin II type 1 receptor autoantibodies; BP, blood pressure; DBP, diastolic blood pressure; P/C, protein-to-creatinine ratio; PIGF, placental growth factor; ROS, reactive oxygen species; sACE2, soluble (or shed) angiotensin-converting enzyme 2; SBP, systolic blood pressure; sFlt-1, soluble fms-like tyrosine kinase-1; and UL, upper limit.

increased metabolic demands of both mother and fetus. Blood volume expands by about 40% to 50%, beginning early in gestation and plateauing in the third trimester.²¹ Glomerular filtration rate increases by 50%; cardiac output rises by ≈20% to 30%, peaking between mid-pregnancy and the third trimester, primarily due to increases in both heart rate and stroke volume.²¹ Systemic vascular resistance decreases markedly, leading to a reduction in arterial pressure despite the elevated cardiac output.²² These hemodynamic changes are essential for ensuring adequate uteroplacental perfusion and fetal development.

Overview of the Renin-Angiotensin-Aldosterone System

The renin-angiotensin-aldosterone system (RAAS) is a cardiovascular hormone system with important roles in regulating blood pressure and fluid homeostasis in pregnant and nonpregnant states. Increased activity of this system is a key component for achieving the physiological adaptations required for a healthy pregnancy. The RAAS is comprised of an ≈60-kD, liver-derived protein, AGT (angiotensinogen), which serves as the only substrate for the rate-limiting, renal-derived enzyme, renin. As depicted in Figure 2, the activity of the RAAS is initiated by the cleavage of a 10-amino acid peptide, Ang (angiotensin) I, from the N terminus of AGT.²³ This highly regulated reaction initiates an enzymatic cascade, whereby Ang I is metabolized into Ang II (the 8-amino acid primary

effector peptide) by ACE (Ang-converting enzyme) and into smaller peptide fragments with varying biological activity by aminopeptidases and carboxypeptidases.²⁴

A primary role for Ang II is stimulation of aldosterone via the AT1R (Ang II type 1 receptor), a G-protein-coupled receptor, in the adrenal cortex. Aldosterone regulates blood volume through sodium and water retention, which is key for promoting the blood volume expansion needed in pregnancy. Ang II exerts other effects, such as vasoconstriction, activation of the sympathetic nervous system, stimulation of ADH (antidiuretic hormone), and promotion of inflammatory factors and reactive oxygen species through AT1R on target sites such as endothelial and smooth muscle cells in blood vessels, brain, heart, and kidney.²⁵ Ang II also signals through the type 2 receptor (AT2R [Ang II type 2 receptor]), which generally opposes the effects of the AT1R (ie, vasodilation, natriuresis, and anti-inflammatory effects) and is mostly expressed during fetal development, but also during certain conditions in adulthood (eg, injury response) and in pregnancy.²⁶

Active Ang II concentration is primarily controlled via negative feedback on renin activity²⁷ but is additionally counter-regulated, especially locally, by metabolism at the C terminus by ACE2 to generate Ang-(1-7), a peptide with purported vasodilatory and other counter-effects to Ang II^{28,29} (Figure 2). Ang-(1-7) can be further metabolized by ACE to Ang-(1-5), which may have cardiovascular and other effects via the AT2R or other G-protein-coupled receptors.³⁰

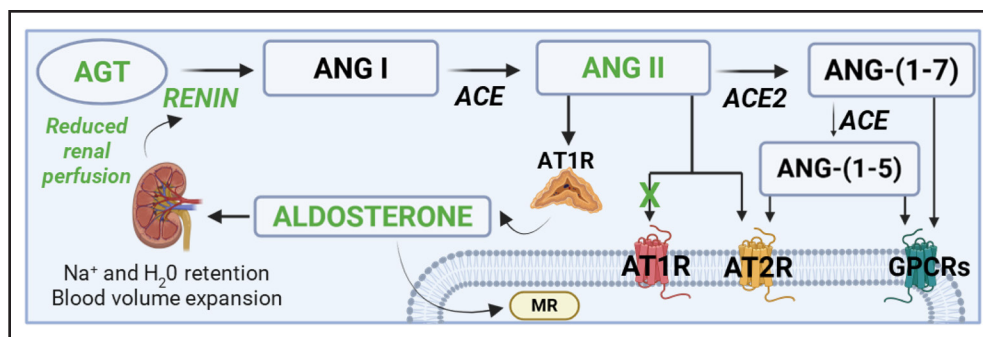


Figure 2. Schematic of the renin-angiotensin-aldosterone system (RAAS).

AGT (angiotensinogen) serves as a substrate for the enzyme renin, resulting in the generation of Ang (angiotensin) I and subsequent formation of Ang II, the main effector hormone of the RAAS. In pregnancy (indicated in green), low renal perfusion pressure stimulates the release of renin from the juxtaglomerular apparatus. Elevated renin activity in combination with a 4-fold increase in AGT (substrate) concentration results in increased formation of Ang peptides and aldosterone. The primary role for Ang II in pregnancy is stimulation of aldosterone release via the AT1R (Ang II type 1 receptor) in the adrenal cortex, leading to sodium and water retention resulting in blood volume expansion. Known effects of AT1R signaling in other organs, such as vasoconstriction, are suppressed, and AT2R (Ang II type 2 receptor) signaling is increased in normal pregnancy. Ang II can be metabolized at the C terminus by ACE2 (Ang-converting enzyme 2) to generate Ang-(1-7) and subsequently Ang-(1-5), where these peptides may exert their effects via MasR (Mas receptor) and other GPCRs (G-protein-coupled receptors). Ang peptides are also metabolized at the n termini by aminopeptidases (not shown). MR indicates mineralocorticoid receptor.

Upregulation of RAAS in Pregnancy

Key components of the RAAS are markedly elevated in human pregnancy. Concentrations of AGT in plasma in nonpregnant states are $\approx 1 \mu\text{mol/L}$ (or $50 \mu\text{g/mL}$) and increase 4- to 5-fold in normotensive women in the third trimester of pregnancy.³¹ The increase in AGT concentration is thought to be mediated by estrogen stimulation of AGT production in the liver.³² In nonpregnant states, plasma renin activity in humans is ≈ 1 to 3 ng/mL per h .³³ In pregnancy, plasma renin activity increases early in the first trimester and continues to rise to a maximum of 3- to 5-fold of nonpregnant activity by the third trimester.^{33,34} Aldosterone concentrations sharply rise after the first trimester, following the upregulation of renin activity, and can reach 8- to 10-fold of nonpregnant levels by the middle of the third trimester.³³ Studies have reported mean blood concentrations of aldosterone in the third trimester exceeding $\geq 800 \text{ pmol/L}$ ^{33,35} compared with 100 to 300 pmol/L in nonpregnant individuals.^{36,37} Limited studies of human pregnancy indicate that plasma renin activity and aldosterone concentrations return to pre-pregnancy values by 6 weeks postpartum³³ in presumably healthy individuals.

Upregulation of RAAS activity plays a vital role in initiating and sustaining key adaptations required for a healthy pregnancy (Figure 3), especially promoting blood volume expansion needed for effective uteroplacental perfusion. Plasma renin activity increases early in the first trimester in a juxtaglomerular response to renal vascular underfilling.³⁸ This results from systemic and renal arterial dilation, via vasodilatory mediators, such as NO, relaxin, progesterone, estrogen,²⁰ and falling peripheral vascular resistance.²¹ Reduced renal perfusion stimulates renin activity and formation of Ang II, which stimulates the release of aldosterone from the adrenal cortex. This

promotes water and salt reabsorption in renal tubules, where the net effect in pregnancy is effective expansion of blood volume (Figure 2).

Preclinical studies suggest a role of the RAAS in the growth and function of the placenta.³⁹ As the interface between mother and fetus, the placenta plays a vital role in determining blood flow and nourishment available to the fetus, which has important consequences on fetal development. The placental RAAS regulates trophoblast proliferation, angiogenesis, and blood flow in the placenta, where AT1R and AT2R are abundantly expressed.^{40,41} In addition, aldosterone is known to have trophic effects in pregnancy via actions at the MR (mineralocorticoid receptor). Mice with a whole body deficiency of aldosterone had reduced litter sizes, fetuses, and placentas.⁴² Treatment of cultured trophoblasts with aldosterone increased proliferation, and blockade of the MR using spironolactone reduced fetal umbilical blood flow in mice.⁴³

RAAS activation supports multiple adaptive organ-level changes during pregnancy. Disruptions in RAAS activity can manifest across organ systems in preeclampsia, including loss of Ang II resistance, inadequate placental development, and systemic vascular dysfunction. Understanding these effects is crucial for distinguishing normal adaptations from preeclampsia pathogenesis.

ACTIVITY OF THE RAAS IN PREECLAMPSIA

In the first stage of preeclampsia, impaired trophoblast invasion and spiral artery remodeling result in insufficient placentation and perfusion of the uteroplacental unit. This results in hypoxia and reperfusion injury, local inflammation and generation of reactive oxygen species, and shedding of antiangiogenic and other particles into

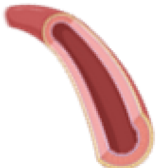
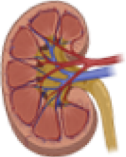
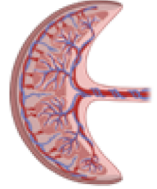
EFFECTS OF PREGNANCY		NT PRE-E	
SYSTEMIC RAAS	AGT, Renin, Ang peptides	↑	↓
	Aldosterone	↑	↓
	sACE2	↑	↑
ORGAN	FEATURE		
 VESSELS	Peripheral Resistance	↓	↑
	Vasodilatory mediators	↑	↓
	Endothelial damage	—	↑
	AT1R signaling	↓	↑
	AT2R expression	↑	↓
	MR signaling	↓	↑
ADRENAL CORTEX 	AT1R → Aldosterone	↑	↓
 KIDNEY	Effective volume expansion	↑	↓
	GFR	↑	↓
	Renin activity	↑	↓
	Arteriole AT1R signaling	↓	↑
	Aldosterone → ENAC	↑	↓
 BRAIN	BBB permeability	—	↑
	CBF autoregulation	—	↓
	AT1R → ADH	↓	↑
	AT1R → SNS	↓	↑
 PLACENTA	AT1R signaling → Vasoconstriction;	↓	↑
	Anti-angiogenic mediators;		
	Inflammation, ROS		
	AT2R expression	↑	↓
	MR → Trophic effects	↑	↓

Figure 3. Systemic activity and organ effects of the renin-angiotensin-aldosterone system (RAAS) in normotensive (NT) and preeclamptic (Pre-E) pregnancy.

Green arrows in NT pregnancy indicate compared with nonpregnant status; Red arrows in Pre-E indicate compared with NT. ADH indicates antidiuretic hormone (or vasopressin); AGT, angiotensinogen; Ang, angiotensin; AT1R, angiotensin II type 1 receptor; AT2R, angiotensin II type 2 receptor; BBB, blood-brain barrier; CBF, cerebral blood flow; ENAC, epithelial sodium channel; GFR, glomerular filtration rate; MR, mineralocorticoid receptor; sACE2, soluble (shed) angiotensin-converting enzyme 2; and SNS, sympathetic nervous system.

the circulation (see Stage 1 in Figure 1). Manifestation of the maternal syndrome is thought to occur as widespread endothelial damage and organ dysfunction in response to these circulating mediators. The maternal syndrome

of preeclampsia is characterized by increasing peripheral resistance, reduced glomerular filtration rate and renal function, ineffective volume expansion, and inadequate blood flow to the uteroplacental unit.

Reduced RAAS Activation in Preeclampsia

In contrast to normotensive pregnancy, where the systemic RAAS is markedly activated, published studies in humans indicate that the activity of the RAAS is reduced in preeclampsia.^{44–47} A 40% reduction in direct renin concentrations and a 4-fold reduction in mean plasma aldosterone concentration were reported in a case-control study of 90 preeclamptic versus 90 uncomplicated pregnancies at 38.5 weeks.⁴⁸ In a prospective study by our laboratory, renin activity and aldosterone concentrations significantly increased from the first (12 weeks) to the third trimester (28 weeks) in 47 women who remained normotensive.⁴⁹ In contrast, in the 27 patients who developed hypertension, renin activity was largely unchanged after the first trimester, and third-trimester aldosterone concentrations were significantly lower than those of the normotensive group. Furthermore, an unchanging or reduction in activity of the RAAS was a key feature among 6 patients who developed preeclampsia compared with 21 patients who developed only gestational hypertension.⁴⁹ These findings are similar to those of prior studies,⁴⁷ including a longitudinal study of 108 women with chronic hypertension where decreasing plasma renin activity over gestation paralleled increasing blood pressure in patients who developed preeclampsia.⁵⁰ The cause of RAAS downregulation in preeclampsia is not well understood but is thought to result from negative feedback on renin activity with increasing peripheral resistance.^{41,50}

Studies also demonstrate disruptions in the alternative arm of the RAAS in normal versus preeclamptic pregnancy, such as reduced concentrations of Ang-(1-7) in preeclampsia.^{34,51} Notably, studies have linked elevated concentrations of circulating (shed) ACE2 with preeclampsia. In a case-control study of preeclampsia with 629 patients, shed or soluble ACE2 concentrations were linked with 60% increased risk for the development of preeclampsia, especially preterm.⁵² ACE2 is a transmembrane protein, where a portion of the extracellular domain can be cleaved by proteases such as ADAM17 (ADAM metallopeptidase domain 17) and released into the circulation. This occurs in response to inflammatory or stress stimuli at local tissue sites and is considered to be a pathological condition. An elevated level of soluble ACE2 in blood suggests local tissue damage and a role for aberrant local RAAS activity in preeclampsia, despite systemic downregulation of the RAAS.

Increased Local AT1R Signaling in Preeclampsia

In normotensive pregnancy, a relative resistance to the pressor effects of Ang II in systemic arterioles occurs to prevent the increase in blood pressure, which would normally occur with activation of the RAAS.⁴⁴ This decrease in vascular sensitivity is lost in preeclampsia, occurring in

a graded fashion over gestation, parallel with the onset of preeclampsia.⁴⁴ This may result from loss of protective factors in patients with preeclampsia who normally prevent AT1R receptor availability during normotensive pregnancy, such as a disrupted endothelial layer, reduced AT2 expression,^{53–56} or lower concentrations of Ang-(1-7).⁵⁷ Thus, despite reduced activation of renin in preeclampsia, concentrations of Ang II at local tissues still bind and induce AT1R signaling. Excessive AT1R signaling likely contributes to multiple organ dysfunction in preeclampsia, such as vasoconstriction in systemic vessels and renal arterioles, and inflammation and production of reactive oxygen species in placental vessels (Figure 3). In addition, some investigators have reported that preeclamptic women harbor antibodies that can bind to the AT1R (AT1R-AA [autoantibodies to the AT1R]) and activate vasoconstrictive, endothelial, and oxidative stress pathways.^{58,59}

Low systemic RAAS status in preeclampsia has been linked with placental-derived antiangiogenic factors. Patients with preeclampsia with an sFlt-1 (soluble fms-like tyrosine kinase-1) to PIGF (placental growth factor) ratio ≥ 85 had low levels of AGT, aldosterone, and plasma renin activity compared with patients with a ratio < 85 .⁴¹ High levels of sFlt-1 increase Ang II sensitivity and hypertension by impairing eNOS (endothelial nitric oxide synthase) phosphorylation and promoting oxidative stress in the vasculature, reducing the bioavailability of the vasodilator, NO.⁶⁰ RAAS also affects PIGF expression, where studies showed that aldosterone can induce the production of PIGF in the placenta.⁶¹ In normotensive pregnancy, vascular response to Ang II is partly blunted by PIGF, suggesting that in preeclampsia, low levels of PIGF contribute to increased sensitivity to Ang II, subsequently hypertension.^{62,63}

Abnormal placental development results in the secretion of placenta-derived exosomes and other placental particles into the maternal circulation.⁶⁴ These shed particles may contain RAAS proteins and peptides, as well as microRNAs targeting renin RNA and autoantibodies to the AT1R, leading to increased sensitivity to Ang II-AT1R signaling in maternal and fetal vessels, and kidney (despite lack of sustained elevation of RAAS), potentially contributing to the maternal syndrome of preeclampsia.⁶⁵ A recent study provided evidence linking placental particles from patients with preeclampsia with vascular dysfunction, where ex vivo incubation of human subcutaneous tissue with placental-derived extracellular vessels from patients with preeclampsia caused increased Ang II-mediated contractility and endothelial damage.⁶⁶

RAAS and Endothelial Dysfunction in Preeclampsia

Evidence from human and animal studies indicates that endothelial dysfunction and vascular injury during

preeclampsia can have long-term effects on maternal vascular health. Mouse models demonstrate that exposure to preeclampsia enhances vascular response to future injuries.⁶⁷ Evidence from human studies indicates the increased sensitivity to Ang II during preeclampsia is present postpartum. A study of 10 women with a history of hypertensive pregnancy compared with 15 normotensive women demonstrated increased salt sensitivity of blood pressure and increased pressor and adrenal responses to infused Ang II during conditions of low-sodium balance.⁶⁸ Another study evaluated Ang II sensitivity and endothelium-dependent dilation in vivo in the microvasculature of 12 women with a history of preeclampsia and 12 control women who had a healthy pregnancy. Preeclampsia was associated with an augmented vasoconstrictor sensitivity to Ang II and reduced endothelium-dependent and nitric oxide-dependent dilation. The latter was augmented with losartan inhibition of the AT1R in patients with previous preeclampsia with no effect in healthy patients.⁶⁹ These data suggest that a loss of protective effects against AT1R signaling during preeclampsia may contribute to microvascular dysfunction and increased risk for long-term vascular disease.

In addition, elevated aldosterone levels are linked to increased cardiac and vascular damage in humans.⁷⁰ Vascular stress and injury increase MR-mediated constriction, inflammation, and fibrosis in vessels (initially an evolutionary survival advantage). In modern conditions, aldosterone excess confers diet- and lifestyle-induced vascular injury,⁷¹ where females may be especially vulnerable. Female sex is primed for increased Ang II stimulation of aldosterone.⁷² A study of human pregnancy by our laboratory demonstrated an association among elevated concentrations of aldosterone in the first trimester and blood pressure, chronic hypertension, and intrauterine growth restriction.⁷³

Preclinical evidence demonstrates that aldosterone has damaging effects on endothelial and smooth muscle cells, which have implications for vascular function during and after pregnancy. Leptin-mediated aldosterone production induced endothelial damage in a study of mouse pregnancy, which was associated with preeclampsia symptoms and reduced fetal weight and placental efficiency.⁷⁴ In nonpregnant mice, aldosterone action on vascular MR enhanced expression of sFlt-1 and augmented vascular wire injury.⁷⁵ Another study demonstrated endothelial MR-mediated diet-induced aortic stiffness in female mice.⁷⁶ In a mouse model of sFlt-1-induced preeclampsia, blood pressure was increased postpartum, and injury resulting from exposure to Ang II was mediated by MRs in smooth muscle cells.⁷⁷ Taken together, data from the literature support RAAS-mediated effects on vasculature as a link between preeclampsia and future maternal vascular diseases.

VASCULAR CONTRIBUTIONS TO COGNITIVE IMPAIRMENT AND DEMENTIA: LINK TO RAAS AND PREECLAMPSIA

Clinical and epidemiological studies have identified preeclampsia as a risk factor for cardiac disease and hypertension.⁷⁸ Recent studies indicate an association of preeclampsia with cognitive decline and dementia later in life. A meta-analysis found that a history of preeclampsia increased the risk of vascular contributions to cognitive impairment and dementia (VCID) by 89%.⁷⁹ VCID is caused by reduced blood flow to the brain, which could result from damage to vessels associated with stroke, small vessel disease, or chronic hypertension. Mechanisms linking hypertension during pregnancy and brain health are not clear. The association between preeclampsia and dementia later in life may be indirectly related to maternal cardiovascular risk, as the risk for cardiovascular disease increases the risk for dementia.⁵⁹ Possible explanations are attributed to shared risk factors between these 2 conditions, which include obesity, insulin resistance, metabolic syndrome, and chronic hypertension (which increases susceptibility to preeclampsia-mediated injuries and further leads to accumulated injuries postpartum and over time). New evidence directly links preeclampsia with increased risk for dementia, where we propose the RAAS as a mechanistic link between these 2 conditions (Figure 4).

Cognitive impairment is linked with cerebral small vessel disease, a condition affecting small blood vessels in the brain and potentially resulting from cerebral hypoperfusion. A recent study of 3623 older adults demonstrated that large fluctuations in cerebral blood flow and increases in cerebrovascular resistance index over time were associated with white matter hyperintensity volume, a key marker for cerebral small vessel disease.⁸⁰ New evidence from a Dutch cohort study of 538 women has linked a history of hypertensive pregnancy with increased white matter hyperintensity volume 15 years after pregnancy.⁸¹ Similar findings were reported from the Hypertensive Disorders of Pregnancy and Brain Health in Midlife CARDIA Study, a prospective cohort of 1441 women assessed with a cognitive battery and a brain magnetic resonance imaging at 25 and 30 years after baseline. Gestational hypertension was associated with impaired cognitive function; greater white matter hyperintensity was observed in the parietal lobes in gestational hypertension and the frontal lobes in preeclampsia.⁸² Direct evidence indicates that these lesions are present during preeclamptic pregnancy. In 94 women with severe preeclampsia, cerebral white matter lesions were evident in 61.7% at delivery, 56.4% at 6 months postpartum, and 47.9% at 1 year postpartum.⁸³ In a retrospective study, cerebral infarcts were identified in 34% of women with eclampsia and 5% of women with preeclampsia.⁸⁴ These data suggest that alterations in

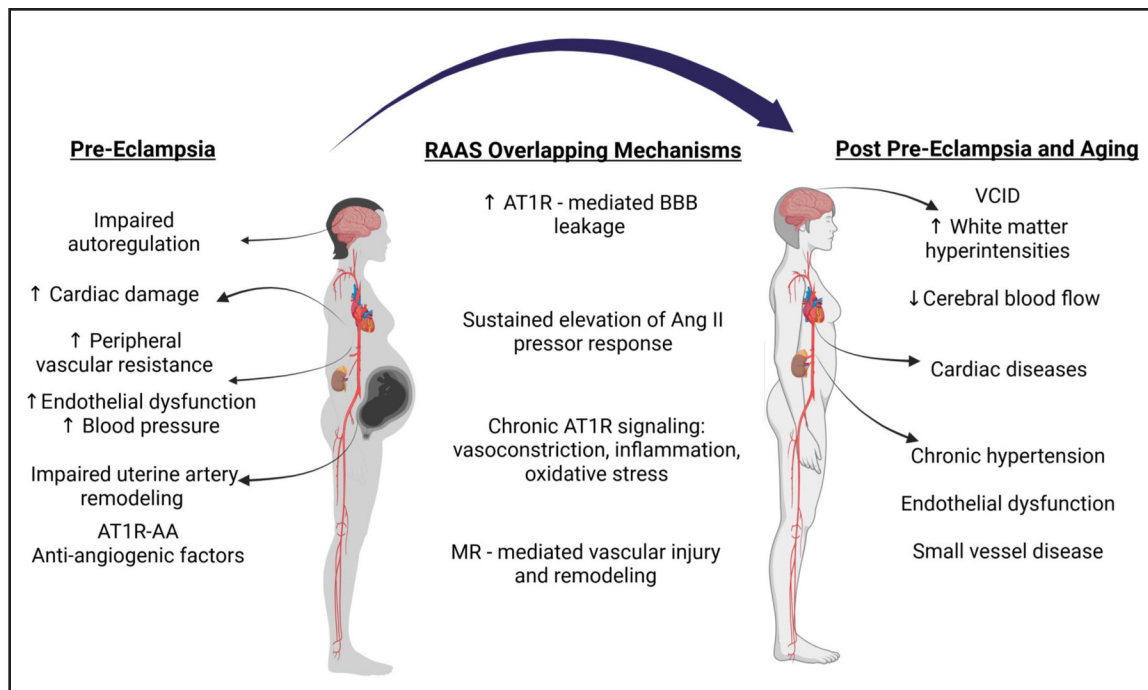


Figure 4. Renin-angiotensin-aldosterone system (RAAS) as a mechanistic link between preeclampsia and vascular contributions to cognitive impairment and dementia (VCID).

Ang II indicates angiotensin II; AT1R, angiotensin II type 1 receptor; AT1R-AA, autoantibodies to the angiotensin II type 1 receptor; BBB, blood-brain barrier; and MR, mineralocorticoid receptor.

cerebral blood flow during hypertensive pregnancy can have negative and long-lasting effects.

Mechanisms of cerebral vessel damage during hypertensive pregnancy include impaired autoregulation,⁸⁵ impaired blood-brain barrier (BBB), and endothelial dysfunction of preeclampsia. Endothelial damage to the BBB has been linked to cognitive decline in older individuals. In a longitudinal study of 295 patients, elevated blood-brain permeability, assessed using cerebrospinal fluid/serum albumin ratio as a marker, was associated with Alzheimer disease.⁸⁶ Several studies have reported evidence of neuroinflammation and BBB disruption in women with preeclampsia and eclampsia,⁸⁷ as well as BBB leakage years after preeclampsia.⁸⁸ Endothelial damage resulting from inflammation and oxidative stress during preeclampsia may lead to increased brain vulnerability to chronic vascular diseases.

Evidence suggests that abnormal activity of the RAAS may contribute to the pathogenesis of small vessel disease and VCID. Ang II modulates BBB permeability via activation in brain endothelial cells⁸⁹; lack of protective effects from AT1R signaling in preeclampsia could contribute to this vulnerability. In addition, decreased activity of ACE and increased activity of ACE2 in frontal white matter were reported in 50 patients with vascular dementia compared with controls, but these effects were not observed in patients with Alzheimer disease.⁹⁰ These data suggest a role for the dysregulated RAAS signaling in the development of cerebral small vessel disease and

VCID. Further studies suggest the damaging effects of aldosterone as a link to VCID. In a rat model of ischemic damage resulting from middle cerebral artery occlusion, MR activation by deoxycorticosterone caused cerebral vessel remodeling and exacerbated the damage caused by cerebral ischemia.⁹¹ Administration of aldosterone to mice increased blood pressure, and increased superoxide production and proinflammatory mRNA expression in cerebral arteries, which was blocked by endothelial-cell deficiency of the MR, though this study used male mice to understand the role of MR in cerebral vascular insult.⁹²

Emerging Interest Areas

Interestingly, sFlt-1, an antiangiogenic tyrosine kinase protein that acts against VEGF (vascular endothelial growth factor), may also be linked with VCID. Levels of sFlt-1, which are released from the placenta, are highly elevated during preeclampsia and can be detected weeks before the development of symptoms associated with preeclampsia.^{93,94} Similarly, altered expression of PIGF can be detected as early as 11 to 13 weeks of gestation; however, its expression is reduced in preeclampsia.⁹⁵ Currently, the sFlt-1/PIGF ratio is used as a screening test for the detection of preeclampsia.⁹⁶

Evidence suggests that antiangiogenic biomarker, sFlt-1, and angiogenic markers, PIGF and VEGF, are associated with markers of cerebral small vessel disease, which is a key contributor to VCID. A recent study

investigated the relationship between cerebrospinal fluid levels of these biomarkers and white matter lesion volume in 742 adults with and without mild cognitive impairment. The findings indicated that higher cerebrospinal fluid levels of PIGF and VEGF-A were significantly associated with greater white matter lesion volume in both cognitively unimpaired individuals and those with mild cognitive impairment.⁹⁷ Lower levels of sFlt-1 were also linked to increased white matter lesion volume in cognitively unimpaired participants, suggesting a complex interplay between proangiogenic and antiangiogenic factors in cerebral vascular health.

Several molecules associated with the RAAS are being considered as potential biomarkers to detect preeclampsia, which includes decreased levels of renin activity, Ang II, aldosterone, and the presence of AT1R autoantibodies.³⁹ However, more translational studies are still needed to determine the specificity and sensitivity of these molecules as ideal biomarkers. In addition, current treatments for preeclampsia do not include anti-RAAS medications such as ACE inhibitors, Ang receptor blockers, and spironolactone due to their teratogenic effects.⁹⁸ Currently, a clinical trial is investigating losartan to decrease subsequent consequences of preeclampsia later in life (Unique identifier: NCT04632589).

An emerging area of interest is exploring the implications of cutaneous RAAS in the contribution to microvascular homeostasis, with potential implications for long-term cerebrovascular health. Data from third-trimester iontophoretic assessments show an augmented hemodynamic response to Ang-(1-7), suggesting that there is dynamic modulation of skin microvascular tone during hypertensive pregnancy.⁵⁷ In patients after preeclampsia, cutaneous administration of Ang-(1-7) revealed restoration of endothelial-dependent microvascular vasodilation, highlighting the loss of vasoprotective activities of Ang-(1-7) as a contributor to vascular dysfunction in hypertensive pregnancy.⁹⁹ In aging populations, Ang II–AT1R–mediated reflex vasoconstriction is amplified, while the counterregulatory AT2R pathway attenuates in older adults^{100,101}; in young females, AT2R functionality supports vasodilation but declines with age.¹⁰² These findings dovetail with the broader conceptual framework of the skin as a regulator of blood pressure through vascular resistance modulation¹⁰³ and RAAS's broader cutaneous roles in vascular integrity and remodeling.¹⁰⁴

Current knowledge of the skin RAAS in pregnancy is limited to small, largely cross-sectional human microvascular studies; longitudinal data linking hypertensive pregnancy to later cerebrovascular disease are lacking. There is a need for prospective cohorts with serial skin microvascular RAAS profiling, from pregnancy through midlife, integrated with systemic RAAS biomarkers, neuroimaging, and cognitive outcomes. Experimental studies should

dissect the mechanistic pathways by which cutaneous Ang-(1-7)/Mas receptor and AT2R signaling modulate systemic vascular resistance and cerebral perfusion, ideally using interventional approaches in both human and translational models.

Summary and Key Messages

Preeclampsia is a dangerous condition of high blood pressure during pregnancy, which increases maternal risk for adverse outcomes during and after pregnancy, including hypertension and vascular dementia later in life. New blood pressure guidelines have linked blood pressure control with risk for dementia, but mechanisms linking these conditions with a history of preeclampsia are not well understood. The RAAS is a key regulator of blood pressure and vascular function, and plays a central role in the cardiovascular adaptation to pregnancy. In this review, we propose that dysregulation of the RAAS in preeclampsia, via loss of sustained renin activation and increased sensitivity to AT1R and MR signaling, contributes to the pathogenesis of preeclampsia and multiple organ dysfunction. Moreover, evidence from animals and humans indicates that vascular and endothelial damage sustained during preeclampsia, as a result of pathogenic AT1R signaling, increases susceptibility to VCID. Future studies may investigate biomarkers of hypertension linking preeclampsia and VCID or define the effects of postpartum RAS antihypertensive therapy on the prevention of hypertension and VCID.

ARTICLE INFORMATION

Affiliations

Department of Neurology (J.L., E.G.S.), Department of Physiology (J.L., K.A.B.), Department of Obstetrics and Gynecology (E.M.), Department of Dietetics and Human Nutrition (E.L.N.), and Department of Pediatrics (R.F.), University of Kentucky, Lexington.

Acknowledgments

BioRender was used to create some figures and images. Finch, R. (2025) <https://BioRender.com/xvm9jj2>.

Sources of Funding

Supported in part by the Regina Drury/Children's Miracle Network Pediatric Research Endowment Fund (RF).

Disclosures

R. Finch serves as a scientific advisor for aTENSION.life, a company making renin-angiotensin-aldosterone system biomarker kits for clinical use. The other authors report no conflicts.

REFERENCES

- Force AT. Hypertension in pregnancy. Report of the American College of Obstetricians and Gynecologists' Task Force on Hypertension in Pregnancy. *Obstet Gynecol*. 2013;122:1122–1131. doi: 10.1097/01.AOG.0000437382.03963.88
- Rana S, Lemoine E, Granger JP, Karumanchi SA. Preeclampsia: pathophysiology, challenges, and perspectives. *Circ Res*. 2019;124:1094–1112. doi: 10.1161/CIRCRESAHA.118.313276

3. LaMarca BB, Bennett WA, Alexander BT, Cockrell K, Granger JP. Hypertension produced by reductions in uterine perfusion in the pregnant rat: role of tumor necrosis factor- α . *Hypertension*. 2005;46:1022–1025. doi: 10.1161/01.HYP.0000175476.26719.36
4. Bakrania BA, Spradley FT, Drummond HA, LaMarca B, Ryan MJ, Granger JP. Preeclampsia: linking placental ischemia with maternal endothelial and vascular dysfunction. *Compr Physiol*. 2020;11:1315–1349. doi: 10.1002/cphy.c200008
5. Conti-Ramsden FI, Nathan HL, De Greeff A, Hall DR, Seed PT, Chappell LC, Shennan AH, Bramham K. Pregnancy-Related acute kidney injury in preeclampsia: risk factors and renal outcomes. *Hypertension*. 2019;74:1144–1151. doi: 10.1161/HYPERTENSIONAHA.119.13089
6. Garovic VD, White WM, Vaughan L, Saiki M, Parashuram S, Garcia-Valencia O, Weissgerber TL, Milic N, Weaver A, Mielke MM. Incidence and long-term outcomes of hypertensive disorders of pregnancy. *J Am Coll Cardiol*. 2020;75:2323–2334. doi: 10.1016/j.jacc.2020.03.028
7. Cameron NA, Petito LC, Shah NS, Perak AM, Catov JM, Bello NA, Capewell S, O'Flaherty M, Lloyd-Jones DM, Greenland P, et al. Association of birth year of pregnant individuals with trends in hypertensive disorders of pregnancy in the United States, 1995–2019. *JAMA Netw Open*. 2022;5:e2228093. doi: 10.1001/jamanetworkopen.2022.28093
8. LeFevre ML; U.S. Preventive Services Task Force. Low-dose aspirin use for the prevention of morbidity and mortality from preeclampsia: U.S. Preventive Services Task Force recommendation statement. *Ann Intern Med*. 2014;161:819–826. doi: 10.7326/M14-1884
9. Lisonkova S, Razaz N, Sabr Y, Muraca GM, Boutin A, Mayer C, Joseph KS, Kramer MS. Maternal risk factors and adverse birth outcomes associated with HELLP syndrome: a population-based study. *BJOG*. 2020;127:1189–1198. doi: 10.1111/1471-0528.16225
10. Al Khalaf S, Khashan AS, Chappell LC, O'Reilly EJ, McCarthy FP. Role of antihypertensive treatment and blood pressure control in the occurrence of adverse pregnancy outcomes: a population-based study of linked electronic health records. *Hypertension*. 2022;79:1548–1558. doi: 10.1161/HYPERTENSIONAHA.122.18920
11. Say L, Chou D, Gemmill A, Tuncalp O, Moller AB, Daniels J, Gulmezoglu AM, Temmerman M, Alkema L. Global causes of maternal death: a WHO systematic analysis. *Lancet Glob Health*. 2014;2:e323–e333. doi: 10.1016/S2214-109X(14)70227-X
12. Callaghan WM, Mackay AP, Berg CJ. Identification of severe maternal morbidity during delivery hospitalizations, United States, 1991–2003. *Am J Obstet Gynecol*. 2008;199:133.e1–133.e8. doi: 10.1016/j.ajog.2007.12.020
13. Mehta A, Spitz J, Sharma S, Bonomo J, Brewer LC, Mehta LS, Sharma G. Addressing social determinants of health in maternal cardiovascular health. *Can J Cardiol*. 2024;40:1031–1042. doi: 10.1016/j.cjca.2024.02.010
14. Lo CCW, Lo ACQ, Leow SH, Fisher G, Corker B, Batho O, Morris B, Chowanec M, Vladutiu CJ, Fraser A, et al. Future cardiovascular disease risk for women with gestational hypertension: a systematic review and meta-analysis. *J Am Heart Assoc*. 2020;9:e013991. doi: 10.1161/JAHA.119.013991
15. Wang IK, Chang SN, Liao CC, Liang CC, Chang CT, Lin HH, Liu JH, Liu YL, Chuang FR, Hsu CY, et al. Hypertensive disorders in pregnancy and preterm delivery and subsequent stroke in Asian women: a retrospective cohort study. *Stroke*. 2011;42:716–721. doi: 10.1161/STROKEAHA.110.594523
16. Lykke JA, Langhoff-Roos J, Sibai BM, Funai EF, Triche EW, Paidas MJ. Hypertensive pregnancy disorders and subsequent cardiovascular morbidity and type 2 diabetes mellitus in the mother. *Hypertension*. 2009;53:944–951. doi: 10.1161/HYPERTENSIONAHA.109.130765
17. Mongraw-Chaffin ML, Cirillo PM, Cohn BA. Preeclampsia and cardiovascular disease death: prospective evidence from the child health and development studies cohort. *Hypertension*. 2010;56:166–171. doi: 10.1161/HYPERTENSIONAHA.110.150078
18. Bushnell C, McCullough LD, Awad IA, Chireau MV, Fedder WN, Furie KL, Howard VJ, Lichtman JH, Lisabeth LD, Pina IL, et al; American Heart Association Stroke Council. Guidelines for the prevention of stroke in women: a statement for healthcare professionals from the American Heart Association/American Stroke Association. *Stroke*. 2014;45:1545–1588. doi: 10.1161/01.str.0000442009.06663.48
19. Garovic VD, Dechend R, Easterling J, Karumanchi SA, McMurtry Baird S, Magee LA, Rana S, Vermut JV, August P; American Heart Association Council on Hypertension; Council on the Kidney in Cardiovascular Disease, Kidney in Heart Disease Science Committee; Council on Arteriosclerosis, Thrombosis and Vascular Biology; Council on Lifestyle and Cardiometabolic Health; Council on Peripheral Vascular Disease; and Stroke Council. Hypertension in pregnancy: diagnosis, blood pressure goals, and pharmacotherapy: a scientific statement from the American Heart Association. *Hypertension*. 2022;79:e21–e41. doi: 10.1161/HYP.0000000000000208
20. Collins HE, Alexander BT, Care AS, Davenport MH, Davidge ST, Eghbali M, Giussani DA, Hoes MF, Julian CG, LaVoie HA, et al. Guidelines for assessing maternal cardiovascular physiology during pregnancy and postpartum. *Am J Physiol Heart Circ Physiol*. 2024;327:H191–H220. doi: 10.1152/ajpheart.00055.2024
21. Sanghavi M, Rutherford JD. Cardiovascular physiology of pregnancy. *Circulation*. 2014;130:1003–1008. doi: 10.1161/CIRCULATIONAHA.114.009029
22. Loerup L, Pullon RM, Birks J, Fleming S, Mackillop LH, Gerry S, Watkinson PJ. Trends of blood pressure and heart rate in normal pregnancies: a systematic review and meta-analysis. *BMC Med*. 2019;17:167. doi: 10.1186/s12916-019-1399-1
23. Tewksbury DA, Frome WL, Dumas ML. Characterization of human angiotensinogen. *J Biol Chem*. 1978;253:3817–3820.
24. Polak Y, Speth RC. Metabolism of angiotensin peptides by angiotensin converting enzyme 2 (ACE2) and analysis of the effect of excess zinc on ACE2 enzymatic activity. *Peptides*. 2021;137:170477. doi: 10.1016/j.peptides.2020.170477
25. Forrester SJ, Booz GW, Sigmund CD, Coffman TM, Kawai T, Rizzo V, Scalia R, Eguchi S. Angiotensin II signal transduction: an update on mechanisms of physiology and pathophysiology. *Physiol Rev*. 2018;98:1627–1738. doi: 10.1152/physrev.00038.2017
26. Carey RM, Wang ZQ, Siragy HM. Role of the angiotensin type 2 receptor in the regulation of blood pressure and renal function. *Hypertension*. 2000;35:155–163. doi: 10.1161/01.hyp.35.1.155
27. Johns DW, Peach MJ, Gomez RA, Inagami T, Carey RM. Angiotensin II regulates renin gene expression. *Am J Physiol*. 1990;259:F882–F887. doi: 10.1152/ajprenal.1990.259.6.F882
28. Vickers C, Hales P, Kaushik V, Dick L, Gavin J, Tang J, Godbout K, Parsons T, Baronas E, Hsieh F, et al. Hydrolysis of biological peptides by human angiotensin-converting enzyme-related carboxypeptidase. *J Biol Chem*. 2002;277:14838–14843. doi: 10.1074/jbc.M200581200
29. Ferrario CM, Chappell MC, Tallant EA, Brosnihan KB, Diz DI. Counterregulatory actions of angiotensin-(1-7). *Hypertension*. 1997;30:535–541. doi: 10.1161/01.hyp.30.3.535
30. Steckelings UM, Widdop RE, Sturrock ED, Lubbe L, Hussain T, Kaschina E, Unger T, Hallberg A, Carey RM, Summers C. The angiotensin AT(2) receptor: from a binding site to a novel therapeutic target. *Pharmacol Rev*. 2022;74:1051–1135. doi: 10.1124/pharmrev.120.000281
31. Tewksbury DA, Dart RA. High molecular weight angiotensinogen levels in hypertensive pregnant women. *Hypertension*. 1982;4:729–734. doi: 10.1161/01.hyp.4.5.729
32. Tewksbury DA, Goodman JR, Kaiser SJ, Burrill RE, Brown HL. Quantitation of the five forms of plasma high molecular weight angiotensinogen in women with pregnancy-induced hypertension. *Am J Hypertens*. 2000;13:221–225. doi: 10.1016/s0895-7061(99)00191-0
33. Langer B, Grima M, Coquard C, Bader AM, Schlaeder G, Imbs JL. Plasma active renin, angiotensin I, and angiotensin II during pregnancy and in preeclampsia. *Obstet Gynecol*. 1998;91:196–202. doi: 10.1016/s0029-7844(97)00660-1
34. Merrill DC, Karoly M, Chen K, Ferrario CM, Brosnihan KB. Angiotensin-(1-7) in normal and preeclamptic pregnancy. *Endocrine*. 2002;18:239–245. doi: 10.1385/ENDO.18.3.239
35. Sanga V, Rossitto G, Seccia TM, Rossi GP. Management and outcomes of primary aldosteronism in pregnancy: a systematic review. *Hypertension*. 2022;79:1912–1921. doi: 10.1161/HYPERTENSIONAHA.121.18858
36. Rehan M, Raizman JE, Cavalier E, Don-Wauchope AC, Holmes DT. Laboratory challenges in primary aldosteronism screening and diagnosis. *Clin Biochem*. 2015;48:377–387. doi: 10.1016/j.clinbiochem.2015.01.003
37. Burrello J, Buffolo F, Domenig O, Tetti M, Pecori A, Monticone S, Poglietti M, Mulatero P. Renin-angiotensin-aldosterone system triple-A analysis for the screening of primary aldosteronism. *Hypertension*. 2020;75:163–172. doi: 10.1161/HYPERTENSIONAHA.119.13772
38. Yamaguchi H, Gomez RA, Sequeira-Lopez MLS. Renin cells, from vascular development to blood pressure sensing. *Hypertension*. 2023;80:1580–1589. doi: 10.1161/HYPERTENSIONAHA.123.00577
39. Alexander BT, South AM, August P, Bertagnoli M, Ferranti EP, Grobe JL, Jones EJ, Loria AS, Safdar B, Sequeira-Lopez MLS; American Heart Association Council on the Kidney in Cardiovascular Disease; Council on Cardiovascular and Stroke Nursing; Council on Cardiovascular Radiology and Intervention; Council on Hypertension; and Council on Lifestyle and Cardiometabolic Health. Appraising the preclinical evidence of the role of the

- renin-angiotensin-aldosterone system in antenatal programming of maternal and offspring cardiovascular health across the life course: moving the field forward: a scientific statement from the American Heart Association. *Hypertension*. 2023;80:e75–e89. doi: 10.1161/HYP.0000000000000227
40. Delforce SJ, Lumbers ER, Morosin SK, Wang Y, Pringle KG. The angiotensin II type 1 receptor mediates the effects of low oxygen on early placental angiogenesis. *Placenta*. 2019;75:54–61. doi: 10.1016/j.placenta.2018.12.001
 41. Verdonk K, Saleh L, Lankhorst S, Smilde JE, van Ingen MM, Garrelds IM, Friesema EC, Russcher H, van den Meiracker AH, Visser W, et al. Association studies suggest a key role for endothelin-1 in the pathogenesis of preeclampsia and the accompanying renin-angiotensin-aldosterone system suppression. *Hypertension*. 2015;65:1316–1323. doi: 10.1161/HYPERTENSIONAHA.115.05267
 42. Todkar A, Di Chiara M, Loffing-Cueni D, Bettoni C, Mohaupt M, Loffing J, Wagner CA. Aldosterone deficiency adversely affects pregnancy outcome in mice. *PLoS Arch*. 2012;464:331–343. doi: 10.1007/s00424-012-1145-4
 43. Gennari-Moser C, Khankin EV, Schüller S, Escher G, Frey BM, Portmann CB, Baumann MU, Lehmann AD, Surbek D, Karumanchi SA, et al. Regulation of placental growth by aldosterone and cortisol. *Endocrinology*. 2011;152:263–271. doi: 10.1210/en.2010-0525
 44. Gant NF, Daley GL, Chand S, Whalley PJ, MacDonald PC. A study of angiotensin II pressor response throughout primigravid pregnancy. *J Clin Invest*. 1973;52:2682–2689. doi: 10.1172/JCI107462
 45. Brown MA, Zammit VC, Mitard DA, Whitworth JA. Renin-aldosterone relationships in pregnancy-induced hypertension. *Am J Hypertens*. 1992;5:366–371. doi: 10.1093/ajh/5.6.366
 46. Tsai YL, Wu SJ, Chen YM, Hsieh BS. Changes in renin activity, aldosterone level and electrolytes in pregnancy-induced hypertension. *J Formos Med Assoc*. 1993;92:514–518.
 47. Elsheikh A, Creatas G, Mastorakos G, Milingos S, Loutradis D, Michalas S. The renin-aldosterone system during normal and hypertensive pregnancy. *Arch Gynecol Obstet*. 2001;264:182–185. doi: 10.1007/s004040000104
 48. Zitouni H, Raguema N, Gannoun MBA, Hebert-Stutter M, Zouari I, Maleh W, Faleh R, Letaifa DB, Almawi WY, Fournier T, et al. Impact of obesity on the association of active renin and plasma aldosterone concentrations, and aldosterone-to-renin ratio with preeclampsia. *Pregnancy Hypertens*. 2018;14:139–144. doi: 10.1016/j.preghy.2018.09.007
 49. Shoemaker R, Poglitsch M, Huang H, Vignes K, Srinivasan A, Cockerham C, Schadler A, Bauer JA, O'Brien JM. Activation of the renin-angiotensin-aldosterone system is attenuated in hypertensive compared with normotensive pregnancy. *Int J Mol Sci*. 2023;24:12728. doi: 10.3390/ijms241612728
 50. Malha L, Sison CP, Helseth G, Sealey JE, August P. Renin-angiotensin-aldosterone profiles in pregnant women with chronic hypertension. *Hypertension*. 2018;72:417–424. doi: 10.1161/HYPERTENSIONAHA.118.10854
 51. Tamanna S, Clifton VL, Rae K, van Helden DF, Lumbers ER, Pringle KG. Angiotensin converting enzyme 2 (ACE2) in pregnancy: preeclampsia and small for gestational age. *Front Physiol*. 2020;11:590787. doi: 10.3389/fphys.2020.590787
 52. Junus K, Björk Ragnarsdóttir I, Nordlöf Callbo P, Bergman L, Lager S, Wikström AK. Elevated mid-pregnancy plasma levels of angiotensin-converting enzyme 2 in women prior to the development of preeclampsia. *Sci Rep*. 2022;12:4109. doi: 10.1038/s41598-022-08081-8
 53. Bakker WW, Henning RH, van Son WJ, van Pampus MG, Aarnoudse JG, Niezen-Koning KE, Borghuis T, Jongman RM, van Goor H, Poelstra K, et al. Vascular contraction and preeclampsia: downregulation of the angiotensin receptor 1 by hemopexin in vitro. *Hypertension*. 2009;53:959–964. doi: 10.1161/HYPERTENSIONAHA.108.127951
 54. Carey LC, Rose JC. The midgestational maternal blood pressure decline is absent in mice lacking expression of the angiotensin II AT2 receptor. *J Renin Angiotensin Aldosterone Syst*. 2011;12:29–35. doi: 10.1177/1470320310376986
 55. Clark CR, Khalil RA. Regulation of vascular angiotensin II type 1 and type 2 receptor and angiotensin-(1-7)/MasR signaling in normal and hypertensive pregnancy. *Biochem Pharmacol*. 2024;220:115963. doi: 10.1016/j.bcp.2023.115963
 56. Quitterer U, Lother H, Abdalla S. AT1 receptor heterodimers and angiotensin II responsiveness in preeclampsia. *Semin Nephrol*. 2004;24:115–119. doi: 10.1016/j.semnephrol.2003.11.007
 57. Yamaleyeva LM, Merrill DC, Ebert TJ, Smith TL, Mertz HL, Brosnihan KB. Hemodynamic responses to angiotensin-(1-7) in women in their third trimester of pregnancy. *Hypertens Pregnancy*. 2014;33:375–388. doi: 10.3109/10641955.2014.911884
 58. Wallukat G, Homuth V, Fischer T, Lindschau C, Horstkamp B, Jupner A, Baur E, Nissen E, Vetter K, Neichel D, et al. Patients with preeclampsia develop agonistic autoantibodies against the angiotensin AT1 receptor. *J Clin Invest*. 1999;103:945–952. doi: 10.1172/JCI4106
 59. Dechend R, Viet D, Müller DN, Ugele B, Brandes RP, Wallukat G, Park JK, Janke J, Barta P, Theuer J, et al. AT1 receptor agonistic antibodies from preeclamptic patients stimulate NADPH oxidase. *Circulation*. 2003;107:1632–1639. doi: 10.1161/01.CIR.0000058200.90059.B1
 60. Burke SD, Zsengeller ZK, Khankin EV, Lo AS, Rajakumar A, DuPont JJ, McCurley A, Moss ME, Zhang D, Clark CD, et al. Soluble fms-like tyrosine kinase 1 promotes angiotensin II sensitivity in preeclampsia. *J Clin Invest*. 2016;126:2561–2574. doi: 10.1172/JCI83918
 61. Eisele N, Albrecht C, Mistry HD, Dick B, Baumann M, Surbek D, Currie G, Delles C, Mohaupt MG, Escher G, et al. Placental expression of the angiogenic placental growth factor is stimulated by both aldosterone and simulated starvation. *Placenta*. 2016;40:18–24. doi: 10.1016/j.placenta.2016.02.004
 62. Zitouni H, Chayeb V, Ben Ali Gannoun M, Raguema N, Bendhafer S, Zouari I, Ben Abdennebi H, Guibourdenche J, Mahjoub T, Gaddour K, et al. Preeclampsia is associated with reduced renin, aldosterone, and PIGF levels, and increased sFlt-1/PIGF ratio, and specific angiotensin-converting enzyme Ins-Del gene variants. *J Reprod Immunol*. 2023;157:103924. doi: 10.1016/j.jri.2023.103924
 63. Espinoza J, Betancourt A, Belfort MA, Shamshirsaz AA, Fox KA, Yallampalli C. Placental growth factor blunts uterine artery responses to angiotensin II. *BJOG*. 2019;126:1058–1064. doi: 10.1111/1471-0528.15656
 64. Morgan HL, Butler E, Ritchie S, Herse F, Dechend R, Beattie E, McBride MW, Graham D. Modeling superimposed preeclampsia using ang II (angiotensin II) infusion in pregnant stroke-prone spontaneously hypertensive rats. *Hypertension*. 2018;72:208–218. doi: 10.1161/HYPERTENSIONAHA.118.10935
 65. Lumbers ER, Delforce SJ, Arthurs AL, Pringle KG. Causes and consequences of the dysregulated maternal renin-angiotensin system in preeclampsia. *Front Endocrinol (Lausanne)*. 2019;10:563. doi: 10.3389/fendo.2019.00563
 66. Erlandsson L, Ohlsson L, Masoumi Z, Rehnström M, Cronqvist T, Edvinsson L, Hansson SR. Preliminary evidence that blocking the uptake of placenta-derived preeclamptic extracellular vesicles protects the vascular endothelium and prevents vasoconstriction. *Sci Rep*. 2023;13:18425. doi: 10.1038/s41598-023-45830-9
 67. Pruthi D, Khankin EV, Blanton RM, Aronovitz M, Burke SD, McCurley A, Karumanchi SA, Jaffe IZ. Exposure to experimental preeclampsia in mice enhances the vascular response to future injury. *Hypertension*. 2015;65:863–870. doi: 10.1161/HYPERTENSIONAHA.114.04971
 68. Saxena AR, Karumanchi SA, Brown NJ, Royle CM, McElrath TF, Seely EW. Increased sensitivity to angiotensin II is present postpartum in women with a history of hypertensive pregnancy. *Hypertension*. 2010;55:1239–1245. doi: 10.1161/HYPERTENSIONAHA.109.147595
 69. Stanhewicz AE, Jandu S, Santhanam L, Alexander LM. Increased angiotensin II sensitivity contributes to microvascular dysfunction in women who have had preeclampsia. *Hypertension*. 2017;70:382–389. doi: 10.1161/HYPERTENSIONAHA.117.09386
 70. Buffolo F, Tetti M, Mulatero P, Monticone S. Aldosterone as a mediator of cardiovascular damage. *Hypertension*. 2022;79:1899–1911. doi: 10.1161/HYPERTENSIONAHA.122.17964
 71. Biwer LA, Wallingford MC, Jaffe IZ. Vascular mineralocorticoid receptor: evolutionary mediator of wound healing turned harmful by our modern lifestyle. *Am J Hypertens*. 2019;32:123–134. doi: 10.1093/ajh/hpy158
 72. Shukri MZ, Tan JW, Manosroi W, Pojoga LH, Rivera A, Williams JS, Seely EW, Adler GK, Jaffe IZ, Karas RH, et al. Biological sex modulates the adrenal and blood pressure responses to angiotensin II. *Hypertension*. 2018;71:1083–1090. doi: 10.1161/HYPERTENSIONAHA.117.11087
 73. Shoemaker R, Poglitsch M, Davis D, Huang H, Schadler A, Patel N, Vignes K, Srinivasan A, Cockerham C, Bauer JA, et al. Association of elevated serum aldosterone concentrations in pregnancy with hypertension. *Biomedicine*. 2023;11:2954. doi: 10.3390/biomed11112954
 74. Faulkner JL, Wright D, Antonova G, Jaffe IZ, Kennard S, Belin de Chantemele EJ. Midgestation leptin infusion induces characteristics of clinical preeclampsia in mice, which is ablated by endothelial mineralocorticoid receptor deletion. *Hypertension*. 2022;79:1536–1547. doi: 10.1161/HYPERTENSIONAHA.121.18832
 75. Jaffe IZ, Newell BG, Aronovitz M, Mohammad NN, McGraw AP, Perreault RE, Carmeliet P, Ehsan A, Mendelsohn ME. Placental growth factor mediates aldosterone-dependent vascular injury in mice. *J Clin Invest*. 2010;120:3891–3900. doi: 10.1172/JCI40205
 76. Jia G, Habibi J, Aroor AR, Martinez-Lemus LA, DeMarco VG, Ramirez-Perez FI, Sun Z, Hayden MR, Meininger GA, Mueller KB, et al. Endothelial mineralocorticoid receptor mediates diet-induced aortic stiffness in females. *Circ Res*. 2016;118:935–943. doi: 10.1161/CIRCRESAHA.115.308269

77. Biwer LA, Lu Q, Ibarrola J, Stepanian A, Man JJ, Carvajal BV, Camarda ND, Zsengeller Z, Skurnik G, Seely EW, et al. Smooth muscle mineralocorticoid receptor promotes hypertension after preeclampsia. *Circ Res*. 2023;132:674–689. doi: 10.1161/CIRCRESAHA.122.321228
78. Wu P, Hatthothuwa R, Kwok CS, Babu A, Kotronias RA, Rushton C, Zaman A, Fryer AA, Kadam U, Chew-Graham CA, et al. Preeclampsia and future cardiovascular health: a systematic review and meta-analysis. *Circ Cardiovasc Qual Outcomes*. 2017;10:e003497. doi: 10.1161/circoutcomes.116.003497
79. Carey C, Mulcahy E, McCarthy FP, Jennings E, Kublickiene K, Khashan A, Barrett P. Hypertensive disorders of pregnancy and the risk of maternal dementia: a systematic review and meta-analysis. *Am J Obstet Gynecol*. 2024;231:196–210. doi: 10.1016/j.ajog.2024.01.013
80. Ma Y, Bos D, Wolters FJ, Niessen W, Hofman A, Ikram MA, Vernooij MW. Changes in cerebral hemodynamics and progression of subclinical vascular brain disease: a population-based cohort study. *Stroke*. 2025;56:95–104. doi: 10.1161/STROKEAHA.124.047593
81. Hussainali RF, Adank MC, Lamballais S, Vernooij MW, Ikram MA, Steegers EAP, Miller EC, Schalekamp-Timmermans S. Hypertension in pregnancy linked to cerebral small vessel disease 15 years later. *Hypertension*. 2025;82:1326–1335. doi: 10.1161/HYPERTENSIONAHA.124.24544
82. Jiang X, Schreiner PJ, Gunderson EP, Yaffe K. Hypertensive disorders of pregnancy and brain health in midlife: the CARDIA study. *Hypertension*. 2024;82:197–205. doi: 10.1161/hypertensionaha.124.22857
83. Soma-Pillay P, Suleman FE, Makin JD, Pattinson RC. Cerebral white matter lesions after pre-eclampsia. *Pregnancy Hypertens*. 2017;8:15–20. doi: 10.1016/j.preghy.2017.02.001
84. Bergman L, Hannsberger D, Schell S, Imberg H, Langenegger E, Moodley A, Pitcher R, Griffith-Richards S, Herroek O, Hastie R, et al. Cerebral infarcts, edema, hypoperfusion, and vasospasm in preeclampsia and eclampsia. *Am J Obstet Gynecol*. 2024;232:550.e1–550.e14. doi: 10.1016/j.ajog.2024.10.034
85. Jones-Muhammad M, Warrington JP. Cerebral blood flow regulation in pregnancy, hypertension, and hypertensive disorders of pregnancy. *Brain Sci*. 2019;9:224. doi: 10.3390/brainsci9090224
86. Tian X, Zhao Y, Zhu Y, Cui M. Association between elevated blood-brain barrier permeability and the risk of progressive cognitive decline: a longitudinal study. *Arch Gerontol Geriatr*. 2024;124:105441. doi: 10.1016/j.archger.2024.105441
87. Bergman L, Hastie R, Zetterberg H, Blennow K, Schell S, Langenegger E, Moodley A, Walker S, Tong S, Cluver C. Evidence of neuroinflammation and blood-brain barrier disruption in women with preeclampsia and eclampsia. *Cells*. 2021;10:3045. doi: 10.3390/cells10113045
88. Canjels LPW, Jansen JFA, Alers RJ, Ghossein-Doha C, van den Kerkhof M, Schiffer V, Mulder E, Gerretsen SC, Aldenkamp AP, Hurks PPM, et al. Blood-brain barrier leakage years after pre-eclampsia: dynamic contrast-enhanced 7-Tesla MRI study. *Ultrasound Obstet Gynecol*. 2022;60:541–548. doi: 10.1002/uog.24930
89. Fleegal-DeMotta MA, Doghu S, Banks WA. Angiotensin II modulates BBB permeability via activation of the AT(1) receptor in brain endothelial cells. *J Cereb Blood Flow Metab*. 2009;29:640–647. doi: 10.1038/jcbfm.2008.158
90. Tayler HM, Skrobot OA, Baron DH, Kehoe PG, Miners JS. Dysregulation of the renin-angiotensin system in vascular dementia. *Brain Pathol*. 2024;34:e13251. doi: 10.1111/bpa.13251
91. Dorrance AM, Rupp NC, Nogueira EF. Mineralocorticoid receptor activation causes cerebral vessel remodeling and exacerbates the damage caused by cerebral ischemia. *Hypertension*. 2006;47:590–595. doi: 10.1161/01.HYP.0000196945.73586.0d
92. Dinh QN, Young MJ, Evans MA, Drummond GR, Sobey CG, Chrissobolis S. Aldosterone-induced oxidative stress and inflammation in the brain are mediated by the endothelial cell mineralocorticoid receptor. *Brain Res*. 2016;1637:146–153. doi: 10.1016/j.brainres.2016.02.034
93. Roberts JM, Rajakumar A. Preeclampsia and soluble fms-like tyrosine kinase 1. *J Clin Endocrinol Metab*. 2009;94:2252–2254. doi: 10.1210/jc.2009-0945
94. Maynard SE, Min JY, Merchan J, Lim KH, Li J, Mondal S, Libermann TA, Morgan JP, Selkoe FW, Stillman IE, et al. Excess placental soluble fms-like tyrosine kinase 1 (sFlt1) may contribute to endothelial dysfunction, hypertension, and proteinuria in preeclampsia. *J Clin Invest*. 2003;111:649–658. doi: 10.1172/JCI117189
95. Poon LC, Zaragoza E, Akolekar R, Anagnostopoulos E, Nicolaides KH. Maternal serum placental growth factor (PlGF) in small for gestational age pregnancy at 11(+0) to 13(+6) weeks of gestation. *Prenat Diagn*. 2008;28:1110–1115. doi: 10.1002/pd.2143
96. Zeisler H, Llurba E, Chantaine F, Vatis M, Staff AC, Sennstrom M, Olovsson M, Brennecke SP, Stepan H, Allegranza D, et al. Predictive value of the sFlt-1:PlGF ratio in women with suspected preeclampsia. *N Engl J Med*. 2016;374:13–22. doi: 10.1056/NEJMoa1414838
97. Gertje EC, Janelidze S, van Westen D, Cullen N, Stomrud E, Palmqvist S, Hansson O, Mattsson-Carlsson N. Associations between CSF markers of inflammation, white matter lesions, and cognitive decline in individuals without dementia. *Neurology*. 2023;100:e1812–e1824. doi: 10.1212/WNL.000000000000207113
98. Sharma DD, Chandresh NR, Javed A, Girgis P, Zeeshan M, Fatima SS, Arab TT, Gopidasan S, Daddala VC, Vaghiasya KV, et al. The management of preeclampsia: a comprehensive review of current practices and future directions. *Cureus*. 2024;16:e51512. doi: 10.7759/cureus.51512
99. Stanhewicz AE, Alexander LM. Local angiotensin-(1-7) administration improves microvascular endothelial function in women who have had preeclampsia. *Am J Physiol Regul Integr Comp Physiol*. 2020;318:R148–R155. doi: 10.1152/ajpregu.00221.2019
100. Lang JA, Kolb KE. Angiotensin II type I receptor blockade attenuates reflex cutaneous vasoconstriction in aged but not young skin. *Am J Physiol Heart Circ Physiol*. 2015;308:H1215–H1220. doi: 10.1152/ajpheart.00017.2015
101. Lang JA, Krajek AC. Age-related differences in the cutaneous vascular response to exogenous angiotensin II. *Am J Physiol Heart Circ Physiol*. 2019;316:H516–H521. doi: 10.1152/ajpheart.00509.2018
102. Turner CG, de Oliveira K, Lu Q, Patel AR, Pulakat L, Jaffe IZ, DuPont JJ. Microvascular angiotensin II type 2 receptor function is enhanced in young females and declines in a model of murine aging. *J Cardiovasc Aging*. 2024;4:19. doi: 10.20517/jca.2024.09
103. Johnson RS, Titze J, Weller R. Cutaneous control of blood pressure. *Curr Opin Nephrol Hypertens*. 2016;25:11–15. doi: 10.1097/MNH.0000000000000188
104. Silva IMS, Assersen KB, Willadsen NN, Jepsen J, Artuc M, Steckelings UM. The role of the renin-angiotensin system in skin physiology and pathophysiology. *Exp Dermatol*. 2020;29:891–901. doi: 10.1111/exd.14159